8.4 Perimeter and Area on the Coordinate Plane

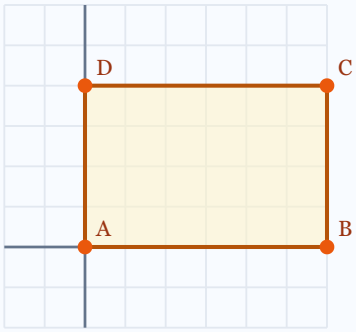
On the coordinate plane, the distance formula gives every side length, so perimeter is just a sum of distances. For area, axis-aligned rectangles and right triangles can be read directly off the grid, and figures with horizontal bases can use the familiar area formulas. For an irregular polygon whose vertices are not conveniently aligned with the axes, use the **Shoelace Formula**: list the vertices in order around the polygon, then $A = \frac{1}{2}\,|\sum(x_i y_{i+1} - x_{i+1} y_i)|$ — multiply each x by the next y, subtract the next x times the current y, add up all these terms, take the absolute value, and halve it. This worksheet builds from basic axis-aligned figures up to the Shoelace Formula for general polygons.

Examples

Example 1: Rectangle perimeter from vertices

Vertices $(0, 0)$, $(6, 0)$, $(6, 4)$, $(0, 4)$. Find the perimeter.

Solution



Sides read directly from the grid: 6 and 4.

Side lengths: 6 and 4 (read directly since the sides are horizontal/vertical).

$P = 2(6 + 4) = 20$.

Answer: 20.

Example 2: Rectangle area from the same vertices

Using the rectangle from Example 1, find the area.

Solution

Identify the shape: the vertices from Example 1 form an axis-aligned rectangle with a horizontal side of length 6 and a vertical side of length 4, so those are the length and width.

Compute: $A = lw = 6(4) = 24$.

Answer: 24 square units.

Example 3: Right triangle area from legs on the axes

Vertices $(0, 0)$, $(8, 0)$, $(0, 6)$. Find the area.

Solution

Identify the legs: one side runs along the x-axis from $(0, 0)$ to $(8, 0)$ (length 8), and the other runs along the y-axis from $(0, 0)$ to $(0, 6)$ (length 6). Since these two sides meet at a right angle at the origin, they are the legs of a right triangle.

Compute the area using $A = \frac{1}{2}(\text{leg}_1)(\text{leg}_2)$: $A = \frac{1}{2}(8)(6) = 24$.

Answer: 24 square units.

Example 4: Distance for one side

Find the length of the side from $(1, 2)$ to $(4, 6)$.

Solution

Identify the tool: with two coordinate points and no shared axis, the distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ is the way to find this side's length.

Compute: $d = \sqrt{(4 - 1)^2 + (6 - 2)^2} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

Answer: 5 units. *Notice this is just a* 3-4-5 *right triangle tilted on the grid.*

Example 5: Right triangle perimeter (Pythagorean triple)

Vertices $(0, 0)$, $(3, 0)$, $(0, 4)$. Find the perimeter.

Solution

Identify the sides: two legs lie along the axes with lengths 3 and 4; the third side (from $(3, 0)$ to $(0, 4)$) is the hypotenuse and must be computed.

Compute the hypotenuse with the Pythagorean Theorem: $\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

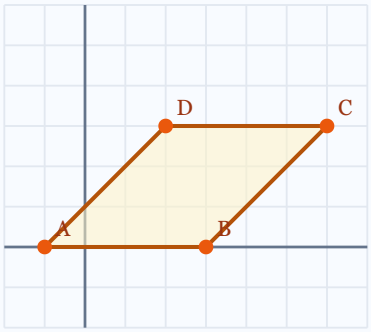
Add all three sides: $P = 3 + 4 + 5 = 12$.

Answer: 12 units.

Example 6: Parallelogram area on the plane

Vertices $A(-1, 0)$, $B(3, 0)$, $C(6, 3)$, $D(2, 3)$. Find the area.

Solution



Base $\overline{AB}$ is horizontal; the height is the vertical gap between $y = 0$ and $y = 3$.

Base $AB = 3 - (-1) = 4$ (horizontal side). Height = vertical distance between the two horizontal sides $= 3 - 0 = 3$.

$A = bh = 4(3) = 12$.

Answer: 12.

Example 7: Square area from vertices

Vertices $(2, 2)$, $(2, 7)$, $(7, 7)$, $(7, 2)$. Find the area.

Solution

Identify the figure: all four sides are horizontal or vertical, and each side has the same length, so this is a square. Read the side length directly from the grid: $7 - 2 = 5$.

Compute: $A = s^2 = 5^2 = 25$.

Answer: 25 square units.

Example 8: Trapezoid area on the plane

Vertices $A(-1, 0)$, $B(3, 0)$, $C(6, 3)$, $D(-4, 3)$. Find the area.

Solution

$\overline{AB}$ (length 4) and $\overline{DC}$ (length 10) are the two horizontal bases. Height = vertical gap = 3.

$$A = \frac{1}{2}(4 + 10)(3) = \frac{1}{2}(14)(3) = 21.$$

Answer: 21.

Example 9: Equilateral triangle — verify with distance, then find perimeter

Vertices $(0, 0)$, $(6, 0)$, $(3, 3\sqrt{3})$. Verify the triangle is equilateral, then find its perimeter.

Solution

$$d_1 = \sqrt{(6 - 0)^2 + (0 - 0)^2} = 6.$$

$$d_2 = \sqrt{(3 - 6)^2 + (3\sqrt{3} - 0)^2} = \sqrt{9 + 27} = \sqrt{36} = 6.$$

$$d_3 = \sqrt{(0 - 3)^2 + (0 - 3\sqrt{3})^2} = \sqrt{9 + 27} = 6.$$

All three sides equal 6, confirming an equilateral triangle. Perimeter = $3(6) = 18$.

Answer: equilateral, confirmed; perimeter = 18.

Example 10: Shoelace Formula — verify against the basic formula

Vertices $(0, 0)$, $(8, 0)$, $(0, 6)$ (same triangle as Example 3). Find the area using the Shoelace Formula and confirm it matches.

Solution

List vertices in order and apply $A = \frac{1}{2}\left|\sum(x_i y_{i+1} - x_{i+1} y_i)\right|$:

$x_1 y_2 - x_2 y_1 = 0(0) - 8(0) = 0$

$x_2 y_3 - x_3 y_2 = 8(6) - 0(0) = 48$

$x_3 y_1 - x_1 y_3 = 0(0) - 0(6) = 0$

Sum $= 0 + 48 + 0 = 48$.

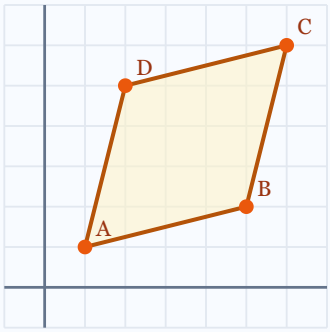
$A = \frac{1}{2}|48| = 24$ — matches Example 3. ✓

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Example 11: Shoelace Formula for an irregular quadrilateral

Vertices (in order) $(1, 1)$, $(5, 2)$, $(6, 6)$, $(2, 5)$. Find the area.

Solution



No horizontal or vertical sides — this needs the Shoelace Formula.

$x_1y_2 - x_2y_1 = 1(2) - 5(1) = 2 - 5 = -3$
$x_2y_3 - x_3y_2 = 5(6) - 6(2) = 30 - 12 = 18$
$x_3y_4 - x_4y_3 = 6(5) - 2(6) = 30 - 12 = 18$
$x_4y_1 - x_1y_4 = 2(1) - 1(5) = 2 - 5 = -3$
$\text{Sum} = -3 + 18 + 18 - 3 = 30.$
$A = \frac{1}{2}|30| = 15.$
Answer: 15 square units.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Perimeter

Rectangle $(0, 0)$, $(5, 0)$, $(5, 3)$, $(0, 3)$. Perimeter?

Answer: **Verification:**

Problem 2: Area

Same rectangle. Area?

Answer: **Verification:**

Problem 3: Triangle

$(0, 0)$, $(10, 0)$, $(0, 4)$. Area?

Answer: **Verification:**

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Problem 4: Perimeter

$(0, 0)$, $(6, 0)$, $(0, 8)$. Perimeter?

Answer: **Verification:**

Problem 5: Square

Side endpoints $(1, 1)$ and $(1, 6)$. Area of the square?

Answer: **Verification:**

Problem 6: Distance

$(2, 3)$ to $(8, 11)$. Length?

Answer: **Verification:**

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Problem 7: Area

Rectangle width 7, height 2. Area?

Answer: **Verification:**

Problem 8: Perimeter

Square side 9. Perimeter?

Answer: **Verification:**

Problem 9: Triangle

(0, 0), (12, 0), (0, 5). Area?

Answer: **Verification:**

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Problem 10: Trapezoid

Bases 6 and 10, height 4. Area?

Answer: **Verification:**

Problem 11: Perimeter

Rhombus side 7. Perimeter?

Answer: **Verification:**

Problem 12: Area

$(1, 1)$, $(7, 1)$, $(7, 4)$, $(1, 4)$. Area?

Answer: **Verification:**

Problem 13: Perimeter

$(0, 0)$, $(5, 0)$, $(5, 12)$. Perimeter of the triangle?

Answer: **Verification:**

Problem 14: Challenge

Use the Shoelace Formula for vertices (in order) $(0, 0)$, $(6, 1)$, $(5, 5)$, $(1, 4)$.

Answer: **Verification:**

Problem 15: Challenge

Vertices $(2, 3)$, $(7, 3)$, $(5, 9)$. Find the area two ways: using base $\times$ height, and using the Shoelace Formula. Confirm they match.

Answer: **Verification:**

✓ Answer Key

1. 16
2. 15
3. 20
4. $6 + 8 + 10 = 24$
5. 25
6. 10
7. 14
8. 36
9. 30
10. 32
11. 28
12. 18
13. $5 + 12 + 13 = 30$
14. 20 square units
15. Both methods give 15 square units

Great work — keep practising!